

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2023

Course: SY. B.Tech. Branch: Artificial Intelligence and Data Science and Allied Semester:III

Subject Code & Name: BTAIC303& Data Structure and Algorithm using Python

Max Marks: 60

Date:14/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Explain the use of the decision control statement. Explain any two decision control statements with suitable examples.	(BT2/CO1)	6
B) What is a dictionary? Explain any 5 operations in the dictionary with a suitable example.	(BT1/CO1)	6
C) Write program to take one list as input and print all even numbers in the input list.	(BT3/CO1)	6
Q.2 Solve Any Two of the following.		
A) What is class and object? Write a program to initialize and display the details of two students using classes and objects.	(BT3/CO2)	6
B) What is a constructor? When constructor is invoked? How it is different from normal methods?	(BT3/CO2)	6
C) Write a code to define a function that will take a string as a parameter and return a reverse string. Call the function with the appropriate parameter.	(BT3/CO2)	6
Q. 3 Solve Any Two of the following.		
A) What is the Linked list? Write the algorithm for inserting a node at the beginning, middle, and end of the doubly linked list.	(BT3/CO3)	6
B) Differentiate between array and linked list? Explain the node structure for singly linked and doubly linked lists with suitable examples.	(BT2/CO3)	6
C) Write a program to perform various operations on the queue.	(BT3/CO1)	6
Q.4 Solve Any Two of the following.		
A) What is Tree? Create a BST for 80,66,48,34,90,80,30,45. After creation of BST perform deletion on node 30.	(BT3/CO4)	6
B) What is Hashing? Explain hashing and collision with suitable examples.	(BT2/CO4)	6
C) What is the heap? Create a min heap for 30,40,20,10,50,60,45,70. Then perform the single delete operation on the heap.	(BT3/CO4)	6
Q. 5 Solve Any Two of the following.		
A) What is Searching? Explain Binary search with a suitable example	(BT3/CO5)	6
B) What is an algorithm? Explain time complexity and space complexity. Analyze the time complexity of insertion sort in the best and worst case.	(BT4/CO5)	6
C) Write a Program for a Linear Search algorithm.	(BT3/CO5)	6

***** End *****